扩展练习

- 来源: https://class.imooc.com/lesson/5259#mid=104964
- 阶段: 阶段一: 【认知基石】理解大模型的本质与工作原理
- 周: 第2周 语言模型怎么训练
- 模块: 监督微调
- 课节: 图文 1-3 监督微调-概念与本质的扩展练习

1-3 监督微调-概念与本质的扩展练习

扩展练习

1

题型: 判断题 难度: 中等

题干: 监督微调和预训练一样，都不需要人工标注数据，可以直接从互联网文本中自动生成。 (复习题)

正确答案: 错误

解析: 两者不同：预训练是自监督学习，答案（下一个Token）在原文中自动获取；监督微调是监督学习，需要人工标注员编写高质量的问答对，每条数据都需要人工设计问题和答案，因此数据量远小于预训练。

2

题型: 阅读理解题 难度: 困难

阅读材料: “Our alignment technique—which we call instruction following—starts with a pre-trained language model and fine-tunes it on a dataset of human-written demonstrations of the desired behavior. […] The key insight is that by training the model to follow instructions across a broad range of tasks, we can make it useful for many applications without task-specific engineering. […] We found that a relatively small number of high-quality demonstrations (around 10,000-50,000 examples) is sufficient to significantly improve instruction following behavior, suggesting that the pre-trained model already possesses much of the knowledge it needs—it just needs to learn how to use it in response to instructions.” （来源：Ouyang et al.,

2022年,《Training Language Models to Follow Instructions with Human Feedback》, InstructGPT论文, Section 2） (延展题)

题型: 单选题 难度: 困难 题号: 1

题干: According to the passage, why is a relatively small number of demonstrations (10,000-50,000) sufficient for instruction following?

选项A: Because the pre-trained model already learns best from small datasets 选项B: Because the pre-trained model already possesses much of the needed knowledge—it just needs to learn how to use it in response to instructions 选项C: Because larger datasets cause catastrophic forgetting 选项D: Because human demonstrations are inherently of higher quality than pre-training data

正确答案: B

解析: The passage explicitly states that “the pre-trained model already possesses much of the knowledge it needs—it just needs to learn how to use it in response to instructions.” This aligns with the chapter’s comparison: pre-training builds a knowledgeable generalist; fine-tuning teaches it how to apply that knowledge to specific tasks.

题型: 简答题 难度: 困难 题号: 2

题干: The authors describe the process as starting with a “pre-trained language model” and fine-tuning it to “follow instructions.” Connect this to Section 2-7’s claim that “监督微调不是重新学习语言，而是在已有能力基础上调整行为模式.” In 1-2 sentences, explain why 10,000-50,000 demonstrations are enough while pre-training needed 300 billion tokens. (延展题)

参考答案: ● Pre-training’s 300 billion tokens build the model’s fundamental language understanding, world knowledge, and reasoning ability from scratch. Fine-tuning’s 10,000-50,000 demonstrations don’t teach new knowledge—they teach a new interaction pattern (question → helpful answer). Since the model already understands language, it only needs to learn when to produce answers versus when to continue text, hence the much smaller data requirement. This is exactly why the chapter says “微调不是重新学习语言，而是在已有能力基础上调整行为模式.”